

# Probability PhD Exam, September 2009

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1. For each fixed  $n$ ,  $X_n^1, X_n^2, \dots, X_n^n$  are independent random variables and have the same Binomial distribution, that is,  $\mathbb{P}(X_n^i = 1) = p_n$  and  $\mathbb{P}(X_n^i = 0) = 1 - p_n$  for  $i = 1, 2, \dots, n$ . Let  $S_n = X_n^1 + X_n^2 + \dots + X_n^n$ . Given the condition that  $np_n$  converges to a positive constant  $\beta$  as  $n$  goes to  $\infty$ , find the limit distribution of  $S_n$  and prove it.
2. Let  $X$  and  $Y$  be two independent random variables with  $\mathbb{E}[Y] = 0$ . Show that for  $p \geq 1$ , we have

$$\mathbb{E}|X + Y|^p \geq \mathbb{E}|X|^p.$$

3. Let  $X$  be a positive random variable with a probability density function  $f(x)$  that is continuous and  $U_n = nX - [nX]$ , where  $[a]$  means the largest integer which is less than  $a$ . Find the limit distribution of  $U_n$  as  $n$  goes to  $\infty$ .
4. Let  $X_1, X_2, \dots, X_n$  be i.i.d. random variables with probability density function  $f(x)$ , and  $M = \max(X_1, X_2, \dots, X_n)$ . Find the conditional expectation  $\mathbb{E}[X_1 | M]$ .
5. For a continuous function  $f(x)$  on  $[0, 1]$ , define

$$P_n(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

Show that  $P_n(x)$  converges to  $f(x)$  on  $[0, 1]$  as  $n$  goes to infinity.

6. Let  $S_n$  be a one-dimensional simple random walk, that is,  $S_n = X_1 + \dots + X_n$ , where  $X_1, \dots, X_n$  are i.i.d. random variables with  $\mathbb{P}(X_1 = +1) = \mathbb{P}(X_1 = -1) = 1/2$ , and  $S_0 = 0$ . and let

$$R_n = 1 + \max_{0 \leq k \leq n} S_k - \min_{0 \leq k \leq n} S_k$$

be the number of points visited by time  $n$ . Prove that  $R_n/\sqrt{n}$  converges weakly to some limit (in distribution).

7. Let  $B_t$  be standard Brownian motion with  $B_0 = 0$  and  $\tau = \inf\{t : B_t = a + bt\}$ , where  $a > 0$  and  $b$  are two constants. Prove that

a).

$$\mathbb{E}_0[e^{-\lambda\tau}] = \exp\{-ab - a\sqrt{b^2 + 2\lambda}\}$$

b).

$$\mathbb{P}_0(\tau < \infty) = \exp\{-2ab\}.$$

8. Let  $\{X_n, n = 0, 1, 2, 3, \dots\}$  be a Markov chain on a countable irreducible state space  $S$ . Let  $\phi$  be a nonnegative function with

$$\mathbb{E}[\phi(X_1) | X_0 = x] \leq \phi(x)$$

for all but a finite number of  $x$ . Prove that if  $\{x | \phi(x) \leq M\}$  is finite for any  $M < \infty$ , then the Markov chain  $\{X_n, n = 0, 1, 2, 3, \dots\}$  is recurrent.